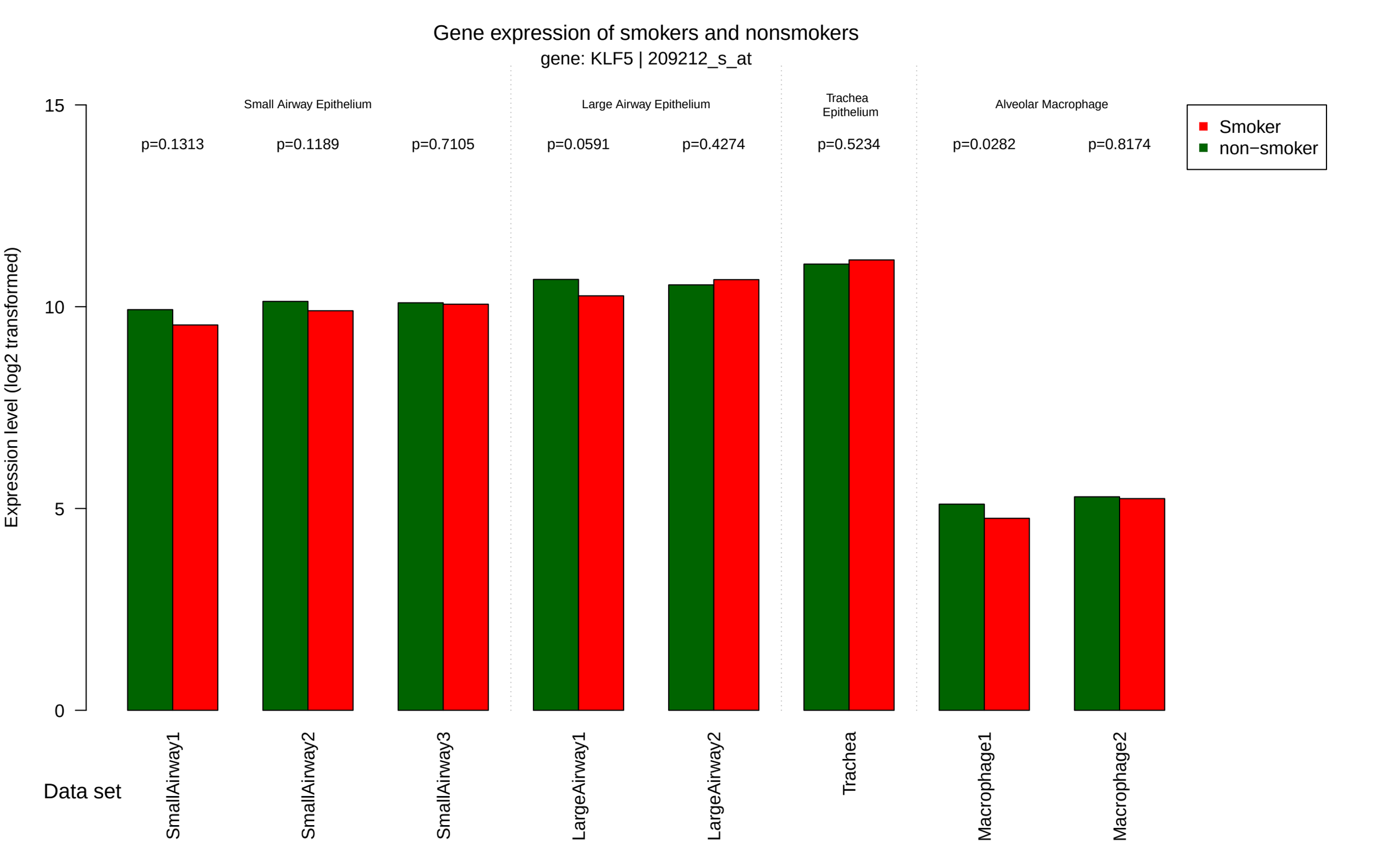
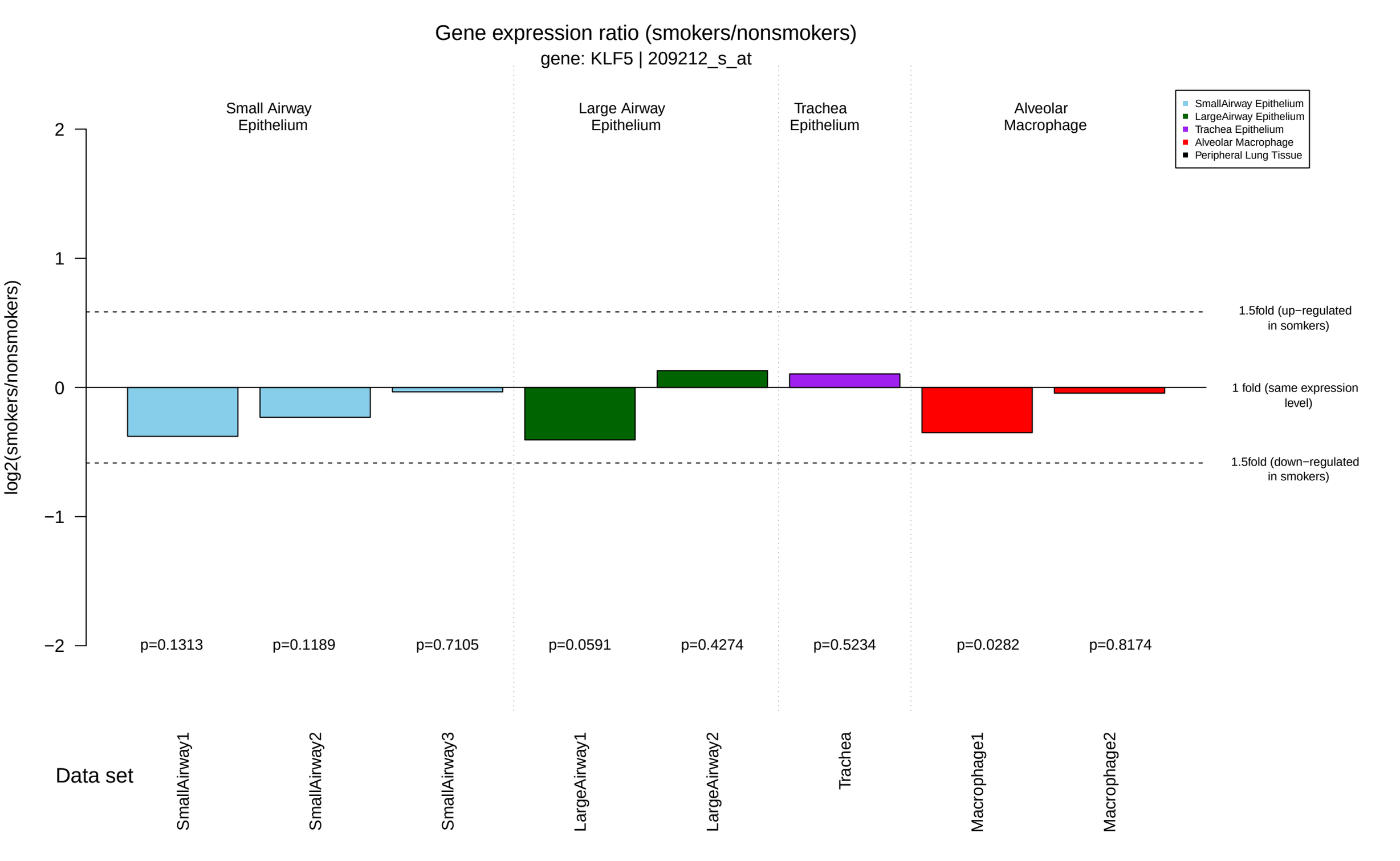
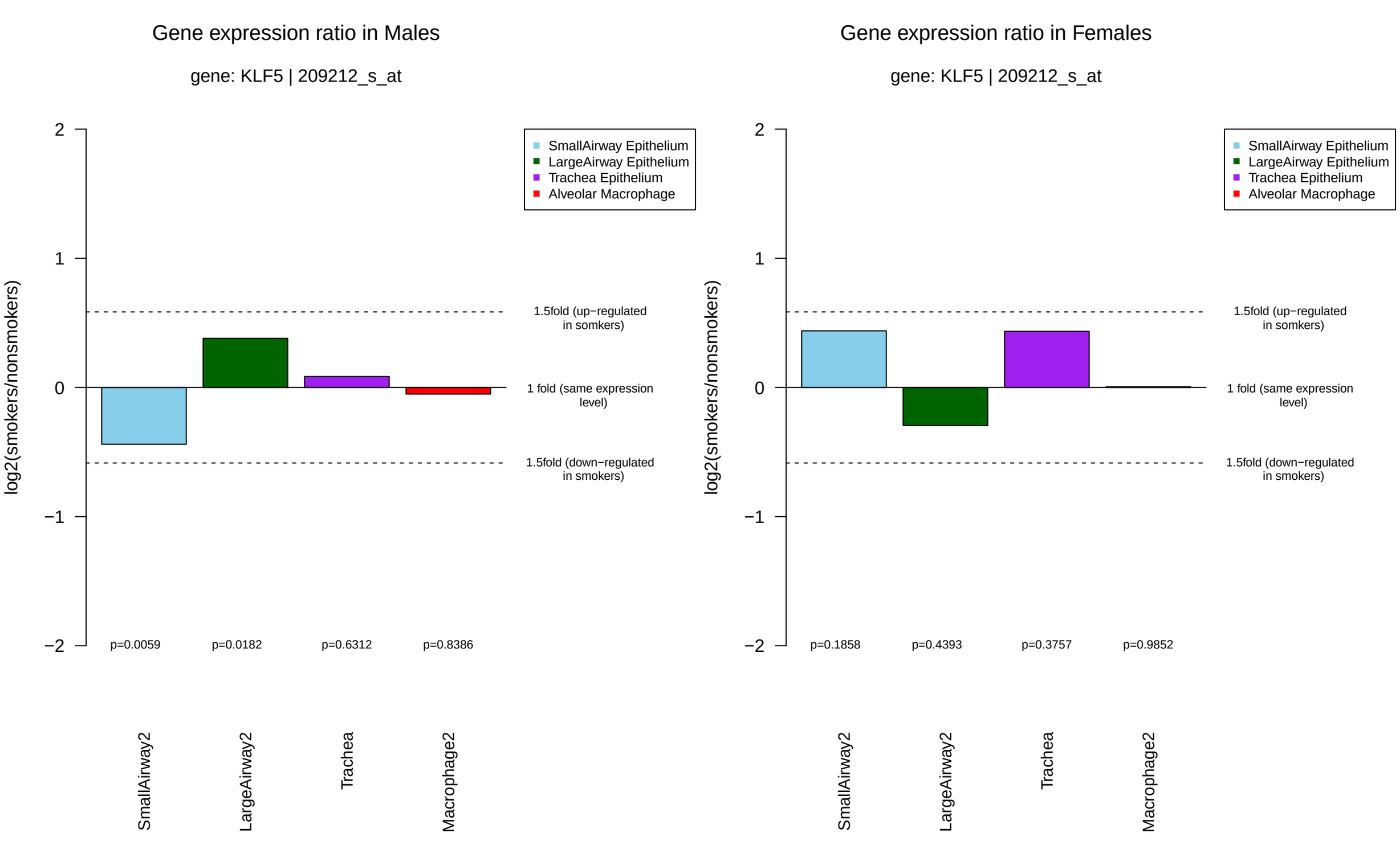
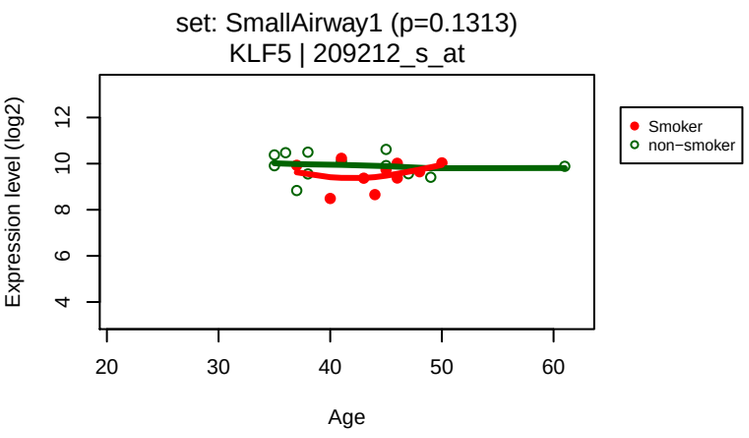
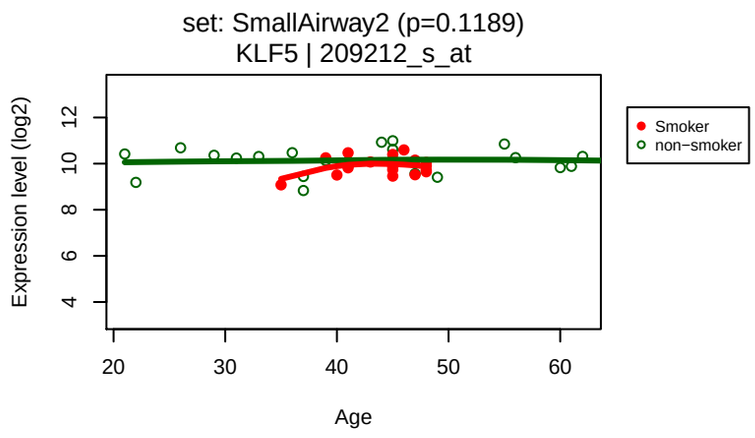
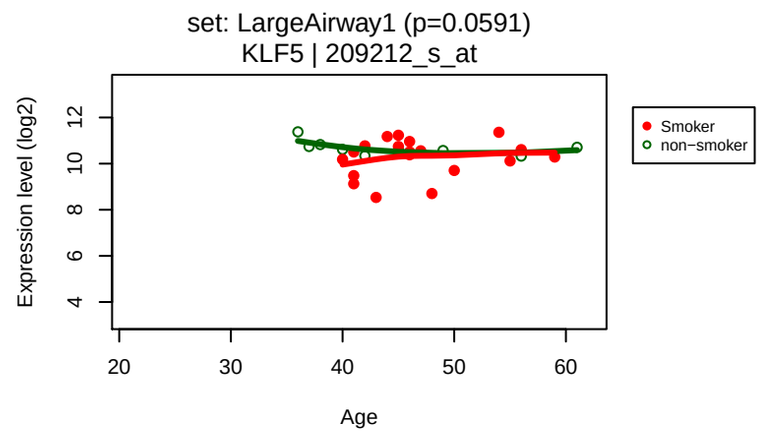

Gene expression of smokers and nonsmokers
gene: KLF5 | 209212_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1313
p=0.1189
p=0.7105
p=0.0591
p=0.4274
p=0.5234
p=0.0282
p=0.8174
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

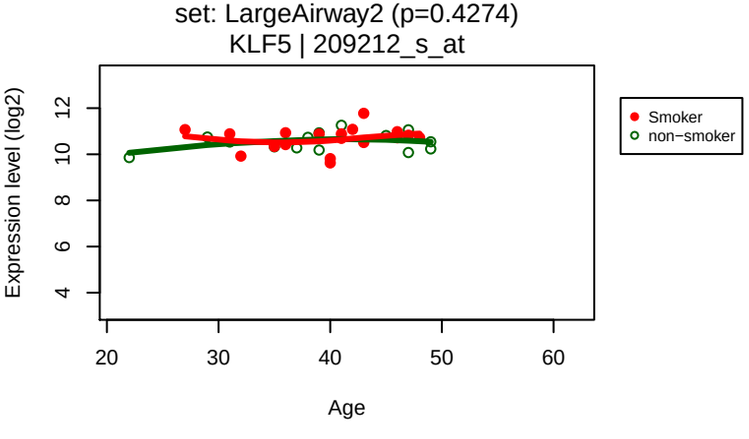

Gene expression ratio (smokers/nonsmokers)
gene: KLF5 | 209212_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1313
p=0.1189
p=0.7105
p=0.0591
p=0.4274
p=0.5234
p=0.0282
p=0.8174
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

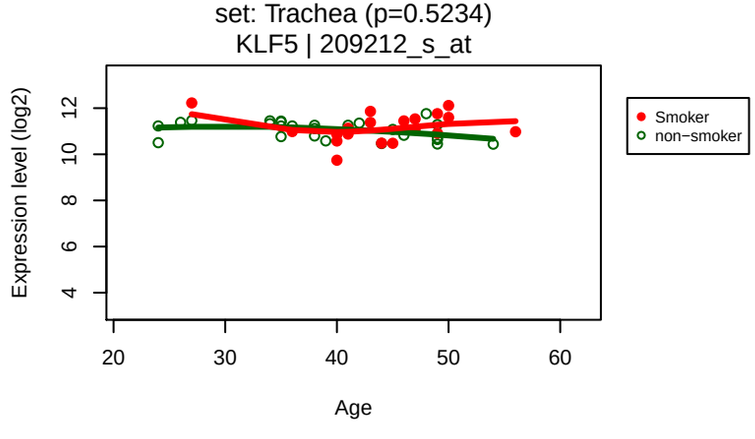

Gene expression ratio in Males
gene: KLF5 | 209212_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0059
p=0.0182
p=0.6312
p=0.8386
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLF5 | 209212_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1858
p=0.4393
p=0.3757
p=0.9852
SmallAirway2
LargeAirway2
Trachea
Macrophage2

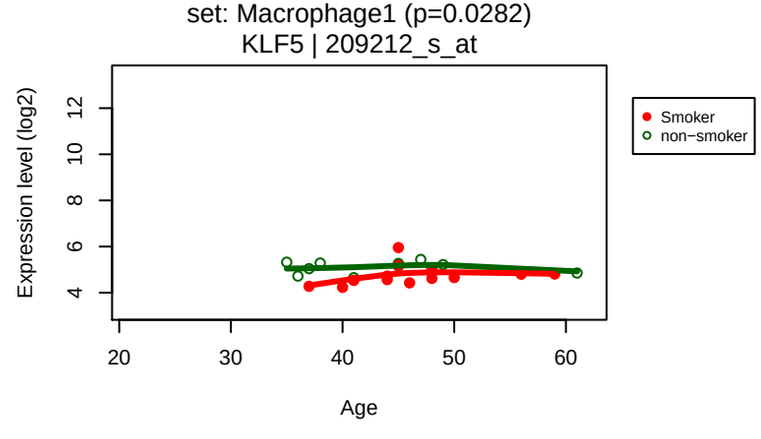

set: SmallAirway1 (p=0.1313)
KLF5 | 209212_s_at
Expression level (log2)
Age
Smoker
non−smoker

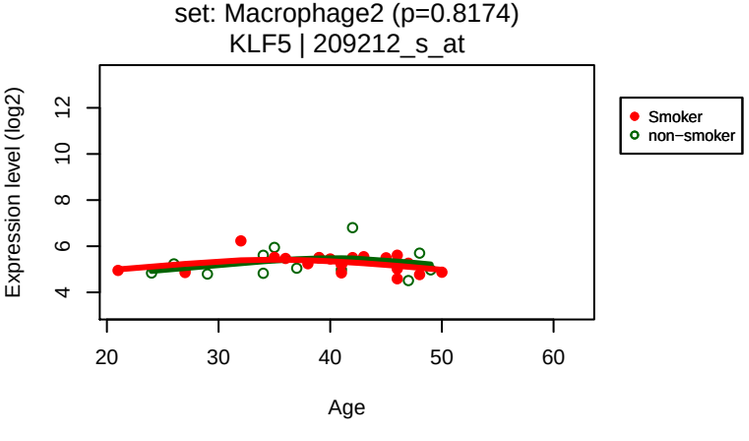

set: SmallAirway2 (p=0.1189)
KLF5 | 209212_s_at
Expression level (log2)
Age
Smoker
non−smoker

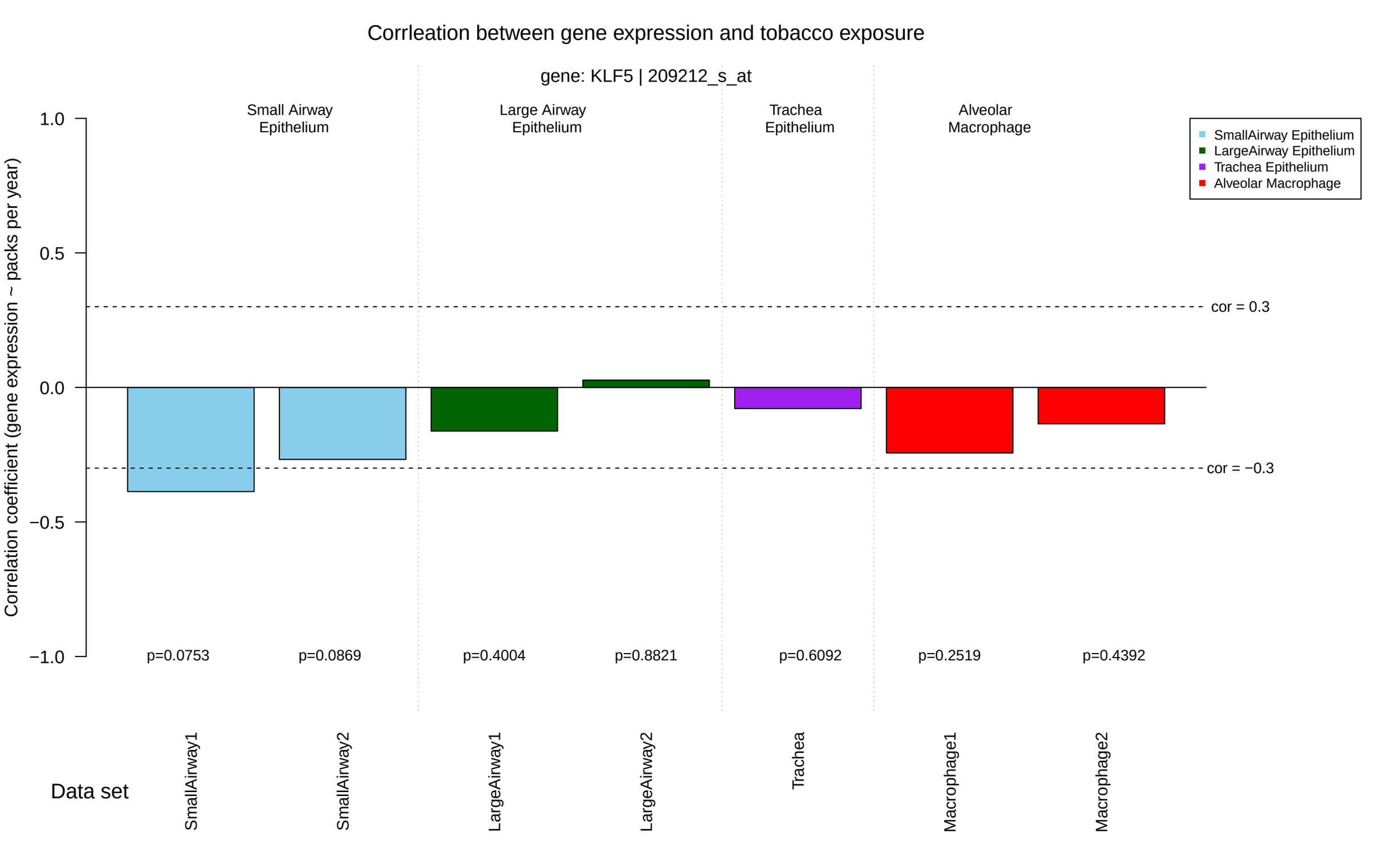

set: LargeAirway1 (p=0.0591)
KLF5 | 209212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4274)
KLF5 | 209212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5234)
KLF5 | 209212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0282)
KLF5 | 209212_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8174)
KLF5 | 209212_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLF5 | 209212_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0753
p=0.0869
p=0.4004
p=0.8821
p=0.6092
p=0.2519
p=0.4392
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2